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The Prevalence, Characteristics and Impact of Chronic Pain in People With Muscular Dystrophies: A Systematic Review and Meta-Analysis

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Abstract: Chronic pain is a frequent, yet under-recognized and under-assessed problem in people with muscular dystrophies (MDs). Knowledge of the prevalence and characteristics of chronic pain, and its impact on function and quality of life is limited and lacks systematic exploration. This article aims to systematically review and synthesize existing literature that addresses chronic pain prevalence, characteristics and impact in people with different types of MDs. The present meta-analysis showed that the estimated prevalence of chronic pain in MDs is high and appears to be similar across different diagnostic groups: 68% (95% CI: 52%−82%) in FSHD, 65% (95% CI: 51%−77%) in DM, 62% (95% CI: 50%−73%) in BMD/DMD, and 60% (95% CI: 48%−73%) in LGMD, although it should be noted that heterogeneity was high in some diagnostic groups. On average, people with FSHD and DM present with moderate pain intensity. The lumbar spine, shoulders and legs are the most frequent sites of chronic pain among people with FSHD, DM, BMD/DMD, and LGMD, with little variation. Diffuse pain across multiple body sites was reported by a notable proportion of these individuals. Chronic pain has a negative impact on daily life activities in people with MDs, and may also contribute to decreased quality of life. The protocol for this review has been published on PROSPERO (CRD42020168096).

Perspectives:* *This is the first systematic review and meta-analysis exploring the prevalence, and nature and impact of chronic pain in people with MDs. The present study demonstrates how common chronic pain is across various MD populations and highlights the need for better recognition and understanding of the nature and impact of pain from health professionals.

Key Words: *Muscular dystrophies, chronic pain, prevalence, systematic review, meta-analysis.*

Abbreviations: *BMD, Becker muscular dystrophy; BPI, brief pain inventory; CIS, checklist individual strength; CMD, congenital muscular dystrophy; CMT, Charcot-Marie-Tooth disorder; DMD, Duchenne muscular dystrophy; DOP, daily observed pain score; DM, myotonic dystrophy; Distal MDs, distal muscular dystrophies; EDMD, Emery-Dreifuss muscular dystrophy; ESS, epworth sleepiness scale; FSHD, facioscapulohumeral muscular dystrophy; HADS, hospital anxiety and depression scale; INQoL, individualized neuromuscular quality of life questionnaire; LGMD, limb-girdle muscular dystrophy; MDs, muscular dystrophies; MPQ, McGill pain questionnaire; MPQ-PPI, present pain intensity in the McGill pain*

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questionnaire; MPQ-PRI, pain rating index in the McGill pain questionnaire; NMDs, neuromuscular diseases; NPRS, numerical pain rating scale; NPS, neuropathic pain scale; OPMD, occulopharyngeal muscular dystrophy; PSG, polysomnographic; PSQI, Pittsburgh sleep quality index; Qi, quality index; SCL, the symptom checklist-90; SF-36, 36-item short-form health survey; UPAT, universal pain assessment tool; VAS, visual analogue scale; WHOQOL-BREF, world health organization quality of life scale brief version.

Introduction

Muscular dystrophies (MDs) are a heterogeneous group of genetic diseases that are among the most common forms of neuromuscular diseases (NMDs). Clinically, MDs manifest as progressive muscle weakness related to loss of mobility, agility and physical movements as a consequence of defects in genes responsible for muscle protein synthesis.[15,30] According to the National Institute of Neurological Disorders and Stroke (NINDS), there are nine major groups of MDs[41]: Duchenne muscular dystrophy (DMD), Becker muscular dystrophy (BMD), Myotonic dystrophy (DM), Facioscapulohumeral muscular dystrophy (FSHD), Congenital muscular dystrophy (CMD), Limb-girdle muscular dystrophy (LGMD), Emery-Dreifuss Muscular Dystrophy (EDMD), Occulopharyngeal Muscular Dystrophy (OPMD), and Distal Muscular Dystrophies (Distal MDs or Distal myopathies). The epidemiology and clinical presentations of the nine major groups of MDs are summarized in Appendix A.

Chronic pain is a common yet frequently under-recognized problem among people with MDs.[16,31,48] While frequently >50%, the prevalence of chronic pain among MDs often differs across studies, which have been performed in different settings, across different types of MDs and with varied methods of data retrieval.[10,22,25,31-33,39,40,42,45,48,51] Similarly, there are variations in other important characteristics such as pain intensity, quality, severity and location.[10,25,31,32,51]

As indicated by many studies,[8,14,34] pain experience interferes with various aspects of people's lives and negatively affects their daily activities, physical and mental health, family and social relationships, and their interaction in the workplace. Furthermore, sleep disturbance is a common complaint in several forms of MDs[7,9] and is known to be reciprocally associated with chronic pain.[19] Despite this, evidence for the impact of MD-related chronic pain on function, sleep, psychological distress and quality of life is limited and lacks systematic exploration.

To date, only one systematic review has assessed the prevalence, characteristics and impact of pain in people with MDs.[11] The authors included 8 studies that assessed pain in DMD and found that pain is a frequent problem in this population, typically occurs in more than one body location and is associated with physical disability and reduced quality of life. This review, however, focused only on DMD and did not clearly focus on 'chronic' pain based on an accepted definition (pain persisting or recurring for a period of 3 months or longer),[49] instead, exploring pain in general, including acute pain that may have been transient in nature. Moreover, the authors did not formally appraise the literature and were therefore unable to report on the strengths and limitations of the evidence included in the review. To the best of our knowledge, no other systematic reviews or meta-analyses have assessed chronic pain in any other type of MD or in MDs more broadly.

Therefore, the aim of this study was to systematically review and synthesize existing literature that addresses chronic pain prevalence, intensity, severity, location, quality, frequency, duration, and its impact in people with different types of MDs. By using a meta-analysis approach, and statistically combining data from individual studies we attempted to provide a more precise estimate of chronic pain prevalence and explore comparisons of pain related outcome measures between different diagnostic groups.

Methodology

This systematic review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[38] A protocol for this review was developed and registered on PROSPERO (CRD: 42020168096).

Search strategy

A systematic search of bibliographic electronic databases, including PubMed, MEDLINE (via EBSCO), CINAHL (via EBSCO), CENTRAL, Scopus, Web of Science, and Allied and Complementary Medicine Database (AMED), was performed for articles (up to March 2020) reporting chronic pain in people with MDs. Medical subject headings and most common key terms searched to identify literature relating to the research question included "muscular dystrophy," "myotonic dystrophy," "Facioscapulohumeral dystrophy," "chronic pain," "persistent pain," and "long-term pain." Key terms differed slightly according to the databases being searched. The complete search strategy for all databases involved in this review is reported in Supplementary Table 1.

Inclusion and exclusion criteria

The identified articles were included if they: 1) involved a population with a diagnosis of any type of MDs, 2) assessed pain in this population, and 3) were published in English. Articles were excluded if they were: 1) not peer-reviewed, 2) performed in a mixed population where independent data on pain in each diagnostic subgroup of MD could not be extracted, or less than ten MD participants were included, 3) not clearly focused on chronic pain (defined as pain persisting or recurring for a period of 3 months or longer[49]; studies that reported data of acute pain or a mixture of acute and chronic pain where independent data of chronic pain could not be extracted were excluded; studies that lacked information on the chronic nature of pain were also excluded).

Study selection

The search strategy was applied to all databases by a reviewer (MH). All the identified studies were downloaded to a reference manager software (EndNote X9) where duplicates were removed. Title and abstract screening were performed independently by 2 reviewers (MH and NM) according to the inclusion and exclusion criteria. For the eligible studies, full-text screening was performed. Disagreements on articles inclusion/exclusion were discussed between the 2 reviewers, and if agreement could not be reached, a third person (DR) was involved. Reference and forward citation searches (via reference lists and Scopus, respectively) were performed for all the articles deemed suitable for inclusion. Full text of the potentially eligible studies identified from the above searches was reviewed to validate the inclusion.

Study quality and risk of bias appraisal

The full text of all included articles was critically appraised by 2 reviewers (MH and NM) independently. Disagreement between reviewers was resolved by discussion and with the involvement of a third reviewer (DR) if necessary.

Quality assessment was conducted using the Risk of Bias Tool for Prevalence Studies.[29] The tool is specifically designed to examine the risk of bias in population-based prevalence studies. It includes 10 items to evaluate the external (items 1−4) and internal (items 5−10) validity and a summary risk of bias assessment. Response options for individual items are either low or high risk of bias, and marked as ‘Yes’ for low risk or ‘No’ for high risk in the scoring sheet. Where not enough information was available on a specific item, high risk of bias was selected.

A summary assessment of the overall risk of study bias at three levels (low, moderate, or high risk of bias) was chosen, based on the reviewer’s subjective judgment given responses to the preceding 10 items in the Risk of Bias Tool for Prevalence Studies. While giving the subjective judgment on the overall risk of study bias, the 2 reviewers agreed that the first 4 items which explore selection bias and nonresponse bias should be assigned more weight. This was because the present systematic review aimed to explore the prevalence of a health issue (chronic pain) in a specific population (people with MDs), where estimates from a biased sample may bring substantial uncertainties to the synthesized findings. Cohen's kappa statistics was used to determine the inter-reviewer agreement of the quality appraisal results.[36]

Data extraction

For each included study, the research design, participant characteristics, and outcome measures were recorded. Data extraction was performed independently by 2 reviewers (MH and NM), and then compared to confirm all extracted data were consistent with the original papers. For studies where outcomes where measured at multiple time points, only baseline data of participants were extracted. Where further information was required, the corresponding authors were contacted by email.

Data analysis

Estimates for the pain prevalence were incorporated into a meta-analysis. Data synthesis was performed by using the MetaXL version 5.3 (EpiGear International, Noosa, Queensland, Australia; an add-in for Microsoft Excel). A double arcsine transformation was used while pooling the prevalence data in order to address the problem of confidence limits outside the 0..1 range and that of variance instability, which arises when the prevalence proportions get close to the limits of the 0..1 range.[3] A quality effects model was used to weight the pooled prevalence data, which was displayed on forest plots with 95% confidence intervals (CIs). Under this model, weights were redistributed due to Qi, a synthetic value computed by dividing each quality score of the studies included in data synthesis by the maximum score in the list of studies.[12,13] Quality scores in this review were referred to the summary assessment of overall risk of bias: low = 3, moderate = 2, and high = 1. Alternative statistical models including the fixed and random effects approaches were simultaneously run while performing data syntheses, and these results can be accessed through Mendeley Data (http://dx.doi.org/10.17632/cfbskh2xn8.2).

A one-way analysis of variance (ANOVA) was used to determine whether there were significant differences on pooled prevalence across different diagnostic groups. The standard deviation (SD) for each group was obtained by the following formula: $SD = \sqrt{N} \times (\text{Upper limit of CI} - \text{Lower limit of CI})/3.92$, N = group sample size, CI = 95% confidence interval.[27] Alternatively, a meta regression via STATA 16.1 was also applied to examine the statistical significance of the group differences.

Pooled average pain intensity estimates were calculated using random effects model meta-analysis. Means and standard deviations were used to summarize the data with mean and 95% CIs. Where number of articles was sufficient, group comparison was performed for studies that reported data for multiple diagnostic groups. While it is recognized that alignment between different pain scales is not always perfect,[53] to allow comparison between different MD groups, the scores of Visual Analogue Scale (VAS) were transformed into the Numerical Pain Rating Scale (NPRS). Since the MetaXL does not support meta-analysis of single means, the ‘meta’ package in R was applied to calculate the overall mean of pain intensity estimates.[44]

Heterogeneity between studies was examined using Higgins's I^2, with values of 25%, 50%, and 75% considered as low, moderate, and high, respectively.[28] Stratified analysis was performed to explore the source of heterogeneity, where sufficient studies allowed this approach. Sensitivity analysis was conducted according to the results of the risk of bias assessment to examine how the quantitative findings of this review were affected by the individual studies with high risks of bias. A narrative synthesis was provided for other outcomes where meta-analysis was not feasible.

Publication bias assessment

We evaluated the possibility of publication bias by visually assessing Doi plots for asymmetry. Doi plots are more sensitive than funnel plots which are hard to interpret when there are less than ten studies.[21] A quantitative measure of the Doi plot asymmetry was performed using the Luis Furuya-Kanamori (LFK) index, where the detected asymmetry can be interpreted into three levels: no asymmetry (LFK index within ±1), minor asymmetry (LFK index exceeds ±1 but within ±2), or major asymmetry (LFK index exceeds ±2).[21] It was hypothesized that, if publication bias was an issue, papers indicating greater prevalence of chronic pain in muscular dystrophies would be more likely to be published. A positive direction of bias for the LFK index was therefore selected a-priori (LFK index > 1).

Findings

Study selection

A total of 609 records were identified through the search. After duplicate removal, 319 studies were deemed suitable for further screening. After title and abstract screening, 74 studies met the inclusion and exclusion criteria. Full-text review identified 11 studies suitable for inclusion. Reference list and forward citation searches of the included studies identified 5 potentially suitable studies of which only one was eligible for inclusion. This led to a total of 12 studies being included in this review. The main reasons for exclusion of articles were the lack of independent data on MDs. We contacted ten authors in an attempt to obtain independent data for individual subgroups which were published as a whole (involving more than one diagnostic subgroup of MDs) or in combination with other populations (eg, other NMDs or physical disabilities). Two responded, but they were not able to provide the

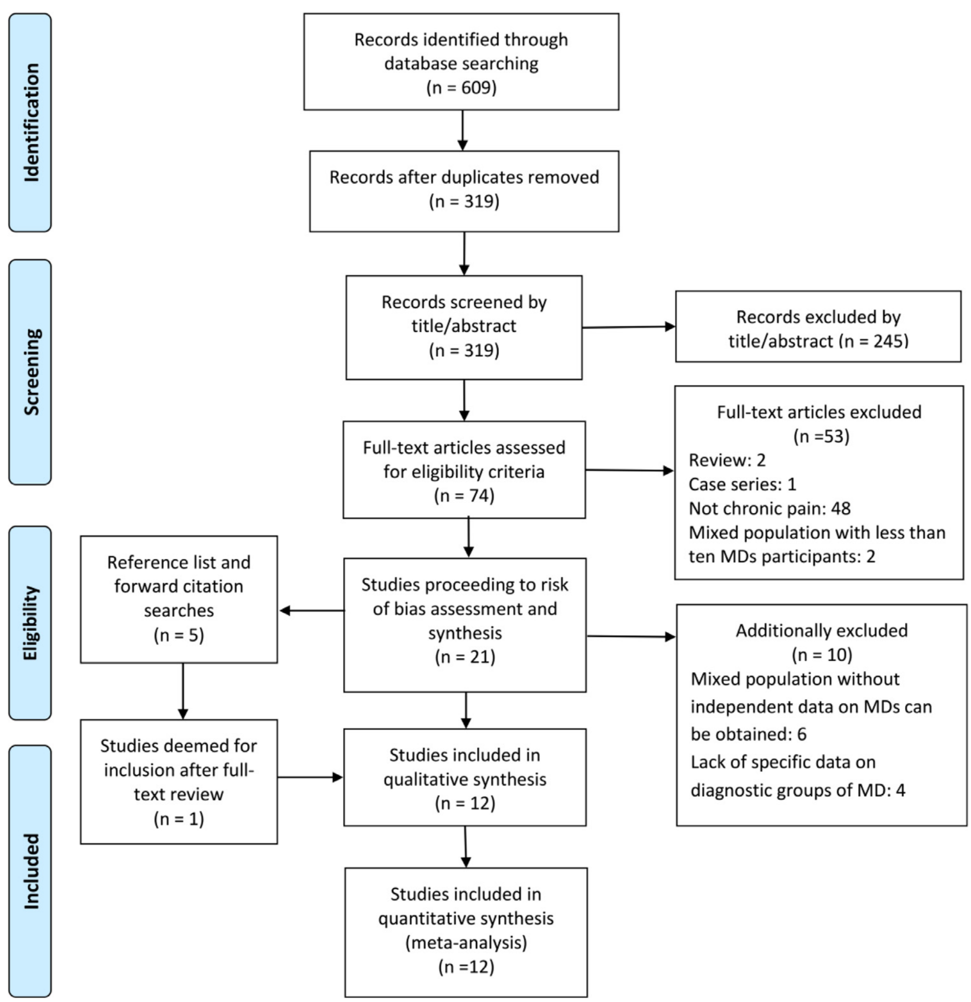

Figure 1. Flow chart of study selection.

Table 1. **Characteristics of Studies**

Study	Study Design	Region	Sample Source	Diagnostic Group	Sample Size	Response Rate	Age (years) Mean (SD)	Gender (Female %)	Primary Aims
Della Marca 2013	Cross-sectional	Italy	An NMD rehabilitation centre	FSHD	55	-	49.6(15.1)	42%	To explore the relationship between sleep disruption and chronic musculoskeletal pain
George 2004	Cross-sectional	The UK	NMD training institution	DM2	24	-	57 (35-69)*	-	To characterize DM2 -associated musculoskeletal pain
Guy-Coichard 2008	Cross-sectional	France	30 centres for NMD consultation	DMD/BMD DM1 FSHD	132 134 121	65%	32.8 (12.5) 46.0 (13.4) 45.7 (14.8)	2% 54% 48%	To assess the characteristics of chronic pain in people with NMDs.
Jensen 2005	Cross-sectional	The USA	NMD rehabilitation clinic and training and research centre	DM FSHD LGMD	26 18 44	47%	51.7 (15.6)	52%	To evaluate the characteristics of chronic pain in people with NMDs.
Jensen 2008	Cross-sectional	The USA	National registry and clinic list	DM FSHD	130 127	78%	46.9 (11.6) 51.8 (13.8)	59% 52%	To evaluate the characteristics of chronic pain in working-aged individuals with DM and FSHD.
Lager 2015	Cross-sectional	Sweden	Swedish national network for NMD	DMD BMD	33 5	71%	15.6 (2.1) 14.2 (2.3)	0 0	To explore the characteristics of chronic pain in adolescents with DMD/BMD and SMA and does the pain differ between diagnoses
Morís 2018	Retrospective	The UK	FSHD Patient Registry	FSHD	398	-	47.0 (60.6)	50%	To examine the characteristics and impact of pain on QoL in people with FSHD.
Moshourab 2016	Cross-sectional	Germany	A Muscle Disorders Outpatient Clinic	DM2	35	-	54.3 (12.7)	54.3%	To analyse the clinical and molecular profile of myalgia in DM2.
Steel 2019	Retrospective	The UK	Tertiary paediatric neuromuscular centre	FSHD	18	-	13.8(3.8)	61%	To explore the clinical course of patients presenting with FSHD in childhood
Tiffreau 2006	Cross-sectional	France	Lille University Medical Centre's PMR clinic	FSHD DM LGMD DMD/BMD CMD	19 15 15 15 4	45%	41 (14.5)	43%	To evaluate the characteristics of chronic pain in people with NMDs.
van der Kooi 2007	RCT	Netherlands	Dutch NMD Association and an NMD training Centre	FSHD	65	-	38(10)	40%	To explore the features of pain and experienced fatigue in people with FSHD, and to study the effects of albuterol and strength training on self-reported pain, experienced fatigue, functional status and psychological distress in these individuals.
Pangalila 2015	Cross-sectional	Netherlands	Multiple home ventilation centres, rehabilitation centres and patient organization for NMDs in Netherlands	DMD	79	53%	28.2 (6.3)	0	To evaluate the frequency of pain, fatigue, anxiety, and depression in adults with DMD and to analyse their relationship with QoL

Abbreviations: NMD, neuromuscular diseases/disorders; FSHD, Facioscapulohumeral muscular dystrophy; DM, Myotonic muscular dystrophy; DMD, Duchenne muscular dystrophy; BMD, Becker muscular dystrophy; LGMD, Limb-girdle muscular dystrophy; CMD, Congenital muscular dystrophy; SMA, Spinal muscular atrophy; QoL, quality of life; RCT, randomized controlled trial.
*Age was reported as Median and range in this study.

necessary data as they did not keep track of specific MD diagnoses in the dataset. The other 8 did not respond within 1 month of the emails being sent and were therefore marked as "no response." Fig 1 shows the study screening and selection processes.

Characteristics of included studies

Characteristics of included studies are presented in Table 1.

Study design

The final 12 articles consisted of 9 cross-sectional studies.[10,22,25,31-33,40,42,48] 2 retrospective studies,[39,45] and one randomized controlled trial (RCT).[51] Of the 9 studies with a cross-sectional design, 6 were surveys,[25,31-33,42,48] with a response rate ranging from 45% to 78%.

Sample sizes varied dramatically across the included articles, ranging from 18 to 398 participants with MDs, with a median participant number of 65. Five studies included a mixed population (one or more diagnostic groups), which consisted of MDs, Spinal muscular atrophy (SMA), Charcot-Marie-Tooth disorder (CMT) and Myasthenia Gravis (MG).[25,31-33,48]

Participants

A total of 1512 participants were included across all the studies, with 6 diagnostic groups of MDs: DM (n = 364), FSHD (n = 821), DMD/BMD (n = 264), LGMD (n = 59), and CMD (n = 4). These participants were largely recruited from Europe (including Italy, the UK, France, Sweden, the Netherlands, and Germany) and North America (the USA). Sources of participants generally included NMD clinics or rehabilitation centers, and national patient registry of NMDs/MDs. The involved participants may be divided into two age groups: adults (18 years of older) and adolescents (less than 18 years). Of the 12 articles included, ten studies investigated chronic pain in adults (mean age ranging from 28 to 57 years), with the remaining two only including teenagers (mean age ranging from 13 to 14 years). The gender distribution across articles tended to be similar, except for the DMD/BMD group, which is an inherited disorder typically affecting males.

Outcomes and measurements

Pain prevalence, pain location, pain intensity and severity, and pain interference were the most common pain outcome measures reported. Other pain domains such as pain quality, frequency, and duration (of episodes) were reported by a few papers only. Most studies utilized validated questionnaires assessing various dimensions of chronic pain. Supplementary Table 2 reports the pain-related outcomes and associated assessment tools utilized by the included studies.

The VAS and NPRS were the most common outcome measures for pain intensity. Present pain intensity index (MPQ-PPI) in the McGill Pain questionnaire (MPQ),[10] Daily Observed Pain score (DOP),[51] and the Universal Pain Assessment Tool (UPAT)[39] were also utilized in some studies for pain intensity evaluation. Pain severity categories were typically determined according to the level of pain intensity assessed by the 0-10 NPRS scale: mild - rated 4 or lower, moderate - rated 5 or 6, severe - rated 7 or higher.[31,32,48] One study specifically asked participants to rate pain severity on a 0 (no pain) to 4 (severe pain) scale.[51]

Body map,[25,48] MPQ,[39,40,51] and Brief Pain inventory (BPI)[31-33] were tools most frequently utilized to record pain location among the included studies. For pain frequency and duration, relevant questions were added to the questionnaire or interview.[22,25] Pain quality was assessed through the MPQ,[22,40] and the Neuropathic Pain Scale (NPS).[31]

Concerning the evaluation of pain interference, most included studies used BPI[25,31-33] to explore the impact of chronic pain on interference in ten different domains (eg, sleep, walking ability, relationships with others). Quality of life was measured by 36-Item Short-Form Health Survey (SF-36),[31] Individualized Neuromuscular Quality of Life Questionnaire (INQoL),[39] and World Health Organization Quality of Life Scale Brief Version (WHOQOL-BREF).[42]

Study quality and risk of bias

Agreement between reviewers on the overall study risk bias assessment was 94%. As depicted in Supplementary Table 3, the risk of bias of the included articles in this systematic review was generally moderate to low. Items 5 to 10 for internal validity (measurement bias and analysis bias) were most frequently met. However, with respect to the external validity (selection bias and nonresponse bias) assessed by items 1 to 4, nearly half of the included articles failed to meet the criteria for item 1, 3, and 4 owing to a biased sample (not a true or close representation of the national population, using inappropriate sampling methods such as a convenience sample) or omitting a comparative analysis of nonresponders.

Key findings

Pain prevalence

While all included studies focused on chronic pain, the reporting period for pain related measures varied across studies. As showed in Table 2, 5 studies investigated pain prevalence over the previous three months.[10,25,31-33] The reporting time period of the remaining studies varied from pain right now,[40] in the past 2 to 4 weeks[42,48,51] or in the past 5 years.[39] Two studies did not provide relevant information regarding pain reporting period.[22,45]

Eight studies (n = 821) presented prevalence data for chronic pain in people with FSHD.[10,25,31,32,39,45,48,51] Reported prevalence of chronic pain in FSHD ranged from 55.6% to 89% (pooled estimate 67.6%; 95% CI:

Table 2. Studies Reporting Estimates for Chronic Pain Prevalence

Study	Reporting Period	FSHD		DM		DMD/BMD		LGMD		Quality Score	Qi
		Participants (N)	Prevalence	Participants (N)	Prevalence	Participants (N)	Prevalence	Participants (N)	Prevalence		
Della Marca 2013	Previous three months	55	76.4%							1	0.33
George 2004	Not mentioned			24	95.8%					1	0.33
Guy-Coichard 2008	Previous three months	121	75.8%	134	66.7%	132	66.4%			3	1
Jensen 2005	Previous three months	18	89%	26	69%			44	64%	2	0.67
Jensen 2008	Previous three months	127	82%	130	60%					3	1
Lager 2015	Previous three months					38	41%			3	1
Morís 2018	Past 5 years	398	55.6%							3	1
Moshourab 2016	Current-point prevalence			35	65.7%					2	0.67
Steel 2019	Not mentioned	18	61.1%							1	0.33
Tiffreau 2006	Previous 3 weeks	19	63%	15	46%	15	67%	15	50%	2	0.67
van der Kooi 2007	Previous 2 weeks	65	80%							2	0.67
Pangalila 2015	Previous 4 weeks					79 (DMD only)	65%			2	0.67

FSHD, Facioscapulohumeral muscular dystrophy; DM, Myotonic muscular dystrophy; DMD, Duchenne muscular dystrophy; BMD, Becker muscular dystrophy; LGMD, Limb-girdle muscular dystrophy.
Qi, a synthetic value computed by dividing each quality score of the studies included in data synthesis by the maximum score in the list of studies.

51.7%−81.9%). Steel, Main, Manzur, Muntoni, Munot[45] was the only study in this diagnostic group to focus on adolescents specifically, and reported a prevalence of 61%, (95% CI: 37%−83%). A forest plot of the studies included in meta-analysis was presented in Fig 2, demonstrating high heterogeneity among the estimates (I^2 87%, $P < .001$). Potential sources of heterogeneity were explored using stratified and sensitivity analyses of the included studies. There was little difference in gender and age distributions between the studies, and thus it was impossible to justify different categories. Five different countries were involved in the eight included studies. Pooling of estimates according to geography suggested that the prevalence of chronic pain in FSHD was higher in the USA (pooled estimate 82%; 95% CI: 76%−88%), compared to that of France (pooled prevalence 75%, 95% CI 63%−85%) and in particular, the UK (pooled estimate 56%; 95% CI: 51%−60%; Fig 3). Differences were also apparent across publication date for studies published within 2000s (pooled estimate 79%; 95% CI: 74%−83%) and studies published within 2010s (pooled estimate 57%, 95% CI: 35%−78%; Fig 4), suggesting a decrease in pain prevalence over time.

Sensitivity analysis suggested that the exclusion of studies with high risk of bias[10,45] did not significantly impact the pooled estimate or heterogeneity. Across all studies, only one provided gender-specific data, with 50.3% males and 61.0% females reporting chronic pain among individuals with FSHD, indicating possible gender differences with respect to chronic pain prevalence.[39]

Six articles (n = 364) specifically examined the prevalence of chronic pain in people with DM.[22,25,31,32,40,48] Prevalence rates ranged widely in this diagnostic group (46%−95.8%). The pooled prevalence estimate is reported in Fig 5 (pooled estimate 65%; 95% CI: 51% −77%), with high heterogeneity (I^2 = 73%, $P < .001$) being detected. Sensitivity analysis showed that the exclusion of one article with high risk of bias[22] dramatically reduced heterogeneity (I2 = 0%, P = .5), but did not impact the pooled prevalence (pooled estimate 63%; 95%CI: 58%−68%).

As presented in Table 2, 4 studies (n = 264) reported chronic pain prevalence in people with DMD and BMD, 3 of them were reported in a combined group (BMD/DMD),[25,33,48] and the other one only included DMD participants.[42] Prevalence rates ranged from 41% to 67% (pooled prevalence 62%, 95% CI 50%−73%; Fig 6), and moderate heterogeneity was observed (I^2 = 63%, P = .04). Lager, Kroksmark[33] was the only study in this diagnostic group to focus on adolescents specifically and reported prevalence of 41%, (95% CI: 26%−51%).

Only 2 studies (n = 59) investigated the prevalence of chronic pain in LGMD.[31,48] The pooled prevalence was similar to the other diagnostic categories (pooled estimate 60%; 95% CI: 48%−73%). As shown in Fig 7, heterogeneity in this meta-analysis was very low (I^2 = 0%, P = .35).

One study involved CMD participants (n = 4), but no independent data on chronic pain prevalence were provided.[48]

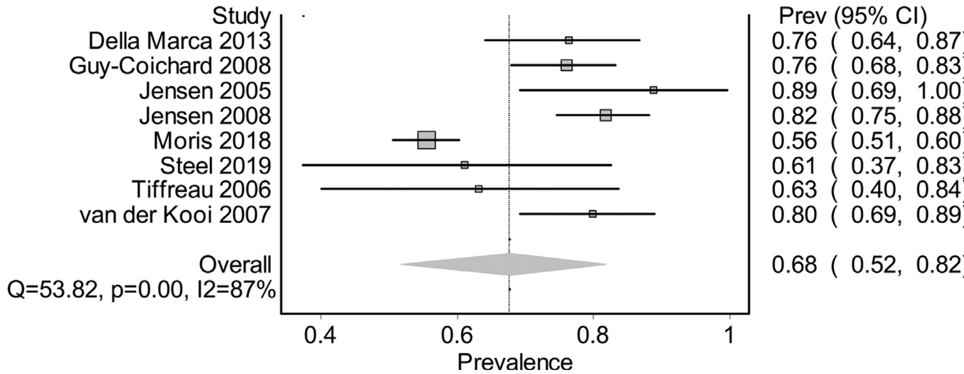

Figure 2. Pooled chronic pain prevalence for FSHD.

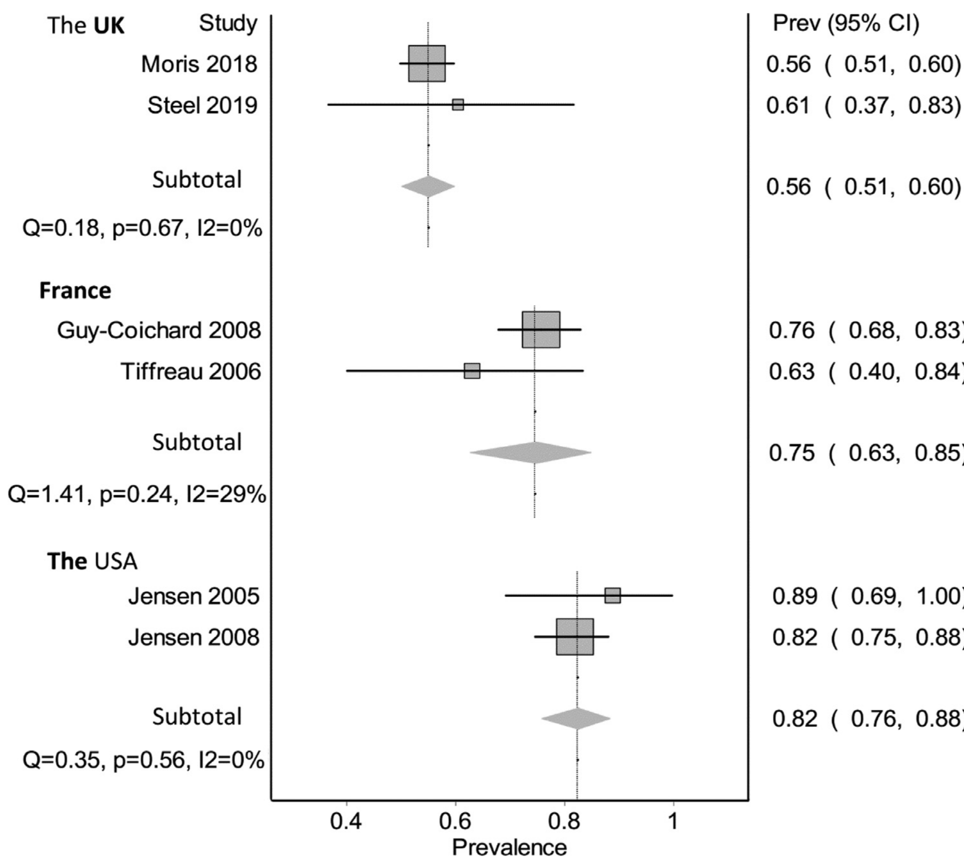

Figure 3. Pooled chronic pain prevalence for FSHD by geography.

There were no statistically significant differences between the pooled prevalence of FSHD, DM, BMD/DMD and LGMD groups as determined by one-way ANOVA ($F_{(3,1504)} = 0.10$, $P = .96$). Likewise, results from meta regression also supported the above findings ($t = -1.27$, $P = .22$, 95%CI: -0.12 to 0.03) (Supplementary Table 6).

Pain intensity

Studies that reported estimates for pain intensity were presented in Table 3. The reporting periods varied across studies, including average pain intensity of right now,[51] in the past week[25,31,32] and in the past three months.[10] One study did not mentioned the associated recall period.[40]

Five studies assessed pain intensity in people with FSHD,[10,25,31,32,51] with average pain intensity estimates ranging widely from 1.6 to 5.1 (pooled mean intensity 4.1, 95% CI 2.6−5.5). Four studies provided average pain intensity data in DM, ranging from 4.2 to 6.3,[25,31,32,40] with a pooled mean intensity of 4.7 (95% CI 4.2–5.2). Forest plots for the above estimates are shown in Fig 8. For the 3 studies that reported average pain intensity data both for individuals with FSHD and DM, a between group comparison was performed. As presented in Table 3, these studies used the same assessment tool (NPRS) and similar reporting periods (7 or 8 days). The meta-analysis results suggested that no significant difference on average pain intensity between the 2 diagnostic groups (Fig 8).

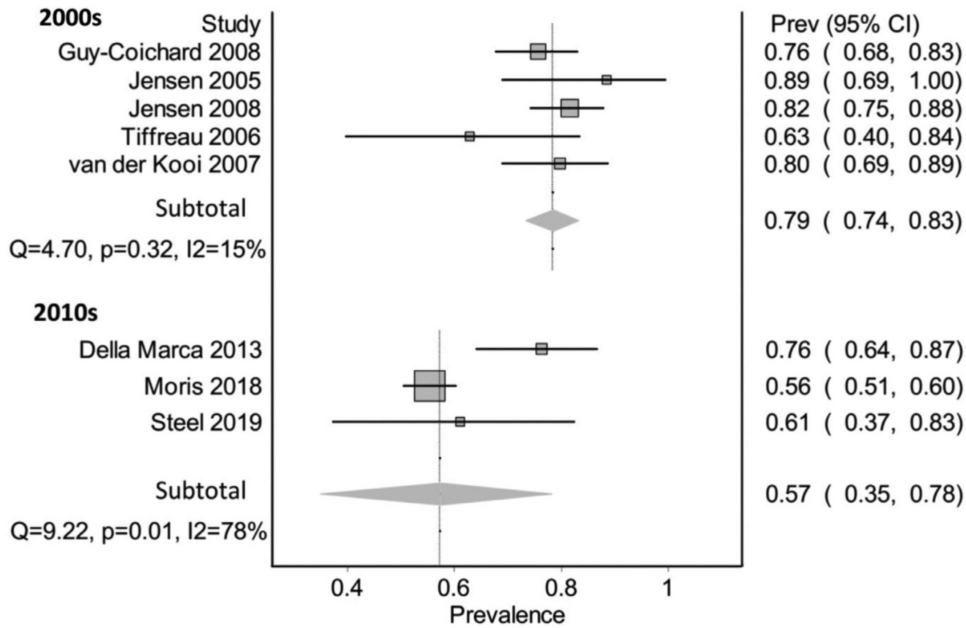

Figure 4. Pooled chronic pain prevalence for FSHD by publication date.

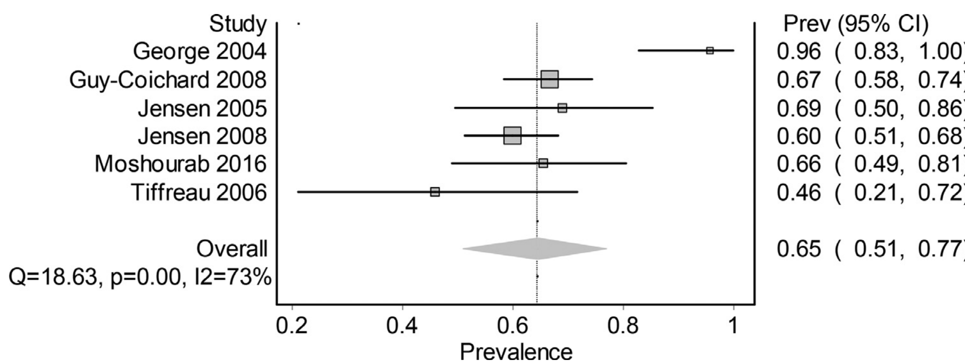

Figure 5. Pooled chronic pain prevalence for DM.

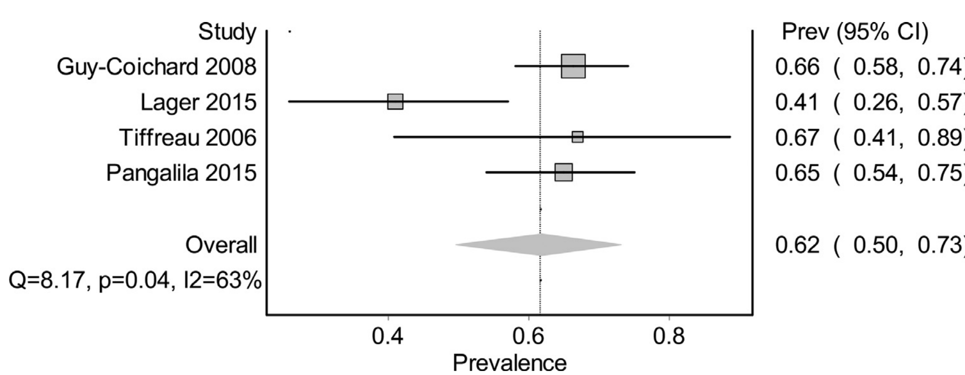

Figure 6. Pooled chronic pain prevalence for BMD/DMD.

Only one study reported data on the average intensity of chronic pain (evaluated by NPRS) in people with BMD/DMD[25] and LGMD,[31] with mean (SD) values of 4.1 (2.3) and 5.25 (2.52), respectively.

Two articles reported pain intensity for the whole sample of people with pain (including acute and chronic pain) and data for the chronic pain subgroup only could not be extracted.[33,48]

Pain severity

Five studies reported pain severity categories among participants,[25,31,32,39,51] with 4 of them providing specific data on the percentages (Supplementary Table 4). The RCT performed by Van der Kooi et al[51] did not provided percentage data, instead only reporting that most participants with FSHD described their pain as mild to

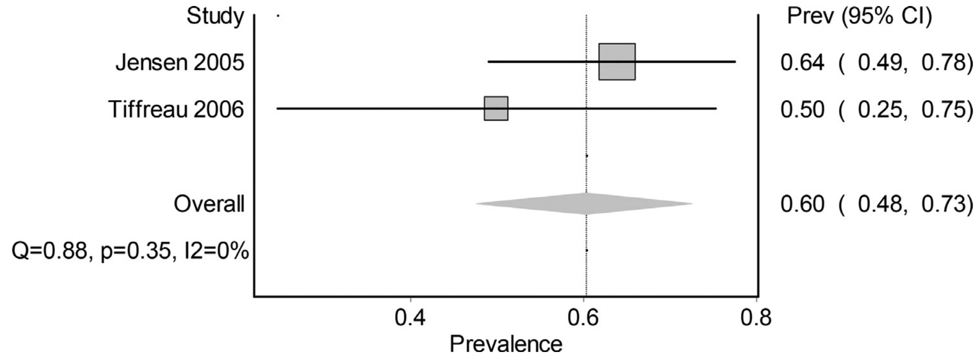

Figure 7. Pooled chronic pain prevalence for LGMD.

moderate when scores on a pain severity scale ranging from 0 (no pain) to 4 (severe pain). As presented in Supplementary table 4, three studies[25,31,32] categorized pain severity based on the pain intensity assessed by a 0 (no pain) to 10 (pain as bad as it can be) NPRS while the remaining article considered the pain severe when reported as horrible or excruciating by the participants.[39]

Within the FSHD diagnostic group, 19% to 21% participants with chronic pain reported their pain as moderate, and 19% to 30% rated their pain as severe.[25,31,32,39] There was greater variability among participants with DM, with 11% to 28% reporting their pain as moderate, and 22% to 50% reporting severe pain.[25,31,32] The distribution of pain severity for the other two diagnostic groups was broadly comparable: BMD/DMD (24% moderate, 19% severe),[25] and LGMD (39% moderate, 25% severe).[31]

Pain location

Eight studies presented the percentages of participants with chronic pain (all FSHD and DM) who reported pain at different locations (as listed in Supplementary Table 5; only studies with diagnosis specific data were presented).[10,22,31,32,39,40,45,51] While some variation was apparent between the two diagnostic groups, the most frequently reported pain sites were similar, and often widespread, including lower back, shoulders, legs, hip, and knees. According to Guy-Coichard et al's[25] findings, diffuse pain sites were seen in 45% of FSHD, 38% of DM, and 36% of BMD/DMD cases. Tiffreau et al[48] revealed an even higher proportion of widespread pain for the whole sample (including DM, FSHD, BMD/DMD, LGMD, and other NMDs; no independent data for each group), with 79% reporting at least 2 pain sites and 63% reporting at least 3 pain sites.

Results from the survey conducted by Jensen et al[31] demonstrated that there were significant differences in terms of the frequency of chronic pain at specific locations in shoulders, hips, and feet. Individuals with FSHD reported pain more frequent in their shoulders and hips and individuals with DM reported pain more frequent in their feet. However, findings from Tiffreau et al[48] suggested that pain locations did not differ between diagnostic groups (including DM, FSHD, BMD/DMD, LGMD, and other NMDs), with the most frequent pain site at the spinal column (81%), followed by shoulder (54%), hip (47%) and knee (47%), similar to the overall data presented in Supplementary Table 5.

Pain quality

Only three studies reported pain quality.[22,31,40] Two of them assessed the pain quality in people with DM by using MPQ pain descriptors.[22,40] As reported by both articles, "tugging," "cramping," "dull," and "tiring" were the descriptors most frequently chosen by the participants with chronic pain. Of interest, a subgroup of individuals described their pain as "radiating" (35%) and "burning" (35%).[40]

Jensen et al[31] used the NPS (a list of 10 descriptors of pain quality) among participants with FSHD, DM, LGMD and other NMDs. Additionally, four descriptors (tiring, sickening, fearful, and punishing) extracted from the SF-MPQ were added to the list to assess the affective component of pain. Their findings suggested that "deep," "tiring," "sharp," and "dull" descriptors were frequently used and were rated (on a 0 to 10 scale) as significantly higher than all the other pain descriptors, which were similar to the results reported by George et al[22] and Moshourab et al.[40] In contrast, pain descriptors that may indicate neuropathic pain, were rated as significantly lower than others. Importantly, according to Jensen et al,[31] it appeared that no pain descriptor was specific to a NMD diagnostic group.

Pain frequency and duration

Pain frequency and duration were reported by only 2 of the included studies.[22,25] Findings from George et al[22] in people with DM 2 (n = 24) suggested that the majority of the participants with chronic pain experienced pain once or several times per week (78%), with nearly half of them experiencing the condition in a frequency of once or several times per day (48%). The duration of episodes varied from seconds to days, with most lasting for hours (87%), followed by seconds to minutes (83%), and days (65%).[22] However, different findings were presented by Guy-Coichard et al[25] who

Table 3. Studies Reporting Estimates for Average Pain Intensity

Study	Measurement	Reporting period	FSHD		DM		DMD/BMD		LGMD	
			Participants (N)	Estimates Mean (SD)	Participants (N)	Estimates	Participants (N)	Estimates	Participants (N)	Estimates
Della Marca 2013	VAS	Previous three months	55	50.9 (27.3)						
Guy-Coichard 2008	NPRS	Previous eight days	121	5.0(2.5)	134	4.6 (2.4)	132	4.1(2.3)		
Jensen 2005	NPRS	Over the past week	18	4.3(2.2)	26	6.3 (3.2)			44	5.3 (2.5)
Jensen 2008	NPRS	Over the past week	127	4.4 (2.4)	130	4.5 (2.8)				
Moshourab 2016	VAS	Not mentioned			35	42.0(22.0)				
van der Kooi 2007	VAS	Pain at the moment	65	16.1 (17.7)						

VAS, Visual analogue scale; NPRS, Numerical Pain Rating Scale; FSHD, Facioscapulohumeral muscular dystrophy; DM, Myotonic muscular dystrophy; DMD, Duchenne muscular dystrophy; BMD, Becker muscular dystrophy; LGMD, Limb-girdle muscular dystrophy.

investigated the characteristics of chronic pain among several MD diagnostic groups: FSHD (n = 121), DM (n = 134), and BMD/DMD (132). No group differences relating to pain frequency and duration were found by diagnoses. For the whole sample, pain occurred daily or almost daily only in 3% of the participants with chronic pain, which was much lower than that reported by George et al.[22] For the duration of pain episodes, 53% episodes were identified as lasting for less than one day, and 47% were lasting for more than 1 day.[25]

Pain interference

Daily life activities. Four studies investigated pain interference on daily life activities by using BPI.[25,31-33] Among adults with FSHD, DM and BMD/DMD, Guy-Coichard et al[25] found that overall pain interference was greatest in FSHD, followed by DM and DMD, while Jensen et al[32] reported no difference between FSHD and DM in overall pain interference. According to Guy-Coichard et al,[25] among people with FSHD, DM, and BMD/DMD, the domains most affected by pain were recreational activities and general activities (including occupational and domestic activities), followed by mobility and mood, while sleep and relationships with others were the least affected. Similarly, recreational activities were found most affected for participants with FSHD, DM, and LGMD[31] and for participants with DM only.[32] However, in the latter study pain was found to interfere most on mobility in participants with FSHD, followed by recreational activities.[32]

Among adolescent boys with BMD/DMD general activity and mood were the domains most affected by pain, followed by mobility and (school) work, with sleep and relationships the least affected.[33]

Quality of life. Three studies explored the relationships between chronic pain and quality of life (QoL).[31,39,42] Participants with FSHD, DM, and LGMD who experienced pain tended to report significantly greater dysfunction in physical functioning, role-physical, bodily pain, general health, vitality, and social functioning than the normative sample of SF-36 in the United States.[31] However, it was unclear whether the dysfunction was primarily caused by chronic pain or the disabling nature of the disorder itself, owing to the lack of comparison between participants with and without pain. Morís et al[39] examined the association of QoL with demographic factors and pain presence in people with FSHD by using multiple regression analysis. Their findings demonstrated that disease duration (Beta = 0.17, P = .003), presence of chronic pain (Beta = 0.12, P = .029) and the total score of MPQ (Beta = 0.4, P = .001) were independently associated with worse QoL. In participants with DMD, those with chronic pain had higher odds of poor physical functioning (OR = 2.75, 95%CI 1.35−5.62, P = .005), but not overall poor QoL, when assessed using univariate logistic regression.[42]

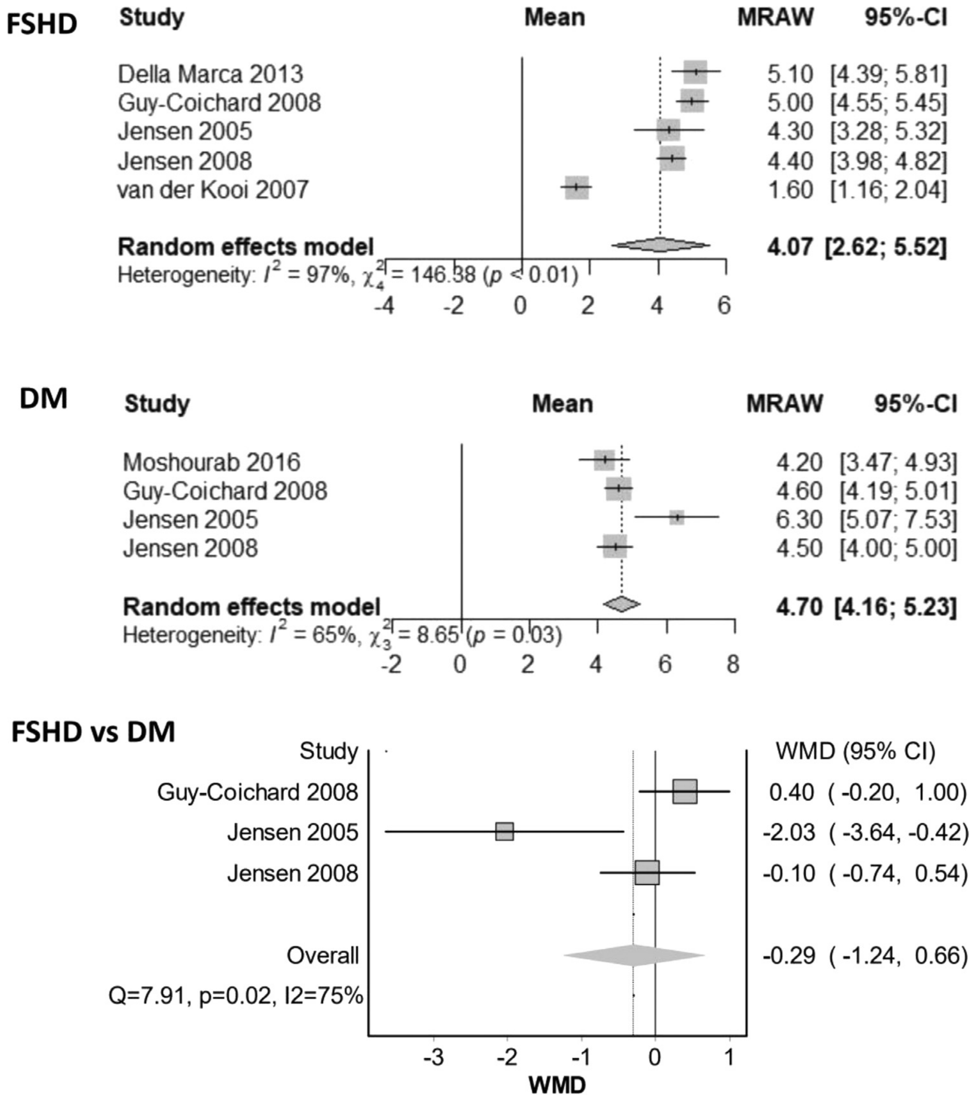

Figure 8. Pooled mean and group comparison of average pain intensity for FSHD and DM.

Publication bias

It was hypothesized that papers indicating greater prevalence of chronic pain in muscular dystrophies would be more likely to be published. A positive direction of bias for the LFK index was therefore selected a-priori (LFK index > 1).

Through visual checking and quantitative measure of Doi plots, major asymmetry were identified for the meta-analysis outcome of pain prevalence in the diagnostic groups of FSHD (LFK index: 4.13; Fig 11), suggesting possible publication bias. No asymmetry was detected for the estimates of DM (LFK index: 0.77) or BMD/DMD (LFK index: −2.24) (Supplementary Fig 1). Doi plot analysis for prevalence in LGMD was impossible owing to the limited number of studies, and the same applied for the pain intensity outcome in FSHD and DM.

Discussion

To our knowledge, this is the first review that systematically explores the prevalence, characteristics and impact of chronic pain in people with MDs. It is also the first to attempt to quantitatively synthesize prevalence data by diagnostic groups in this population. Based on our findings, moderate to high quality evidence shows that chronic pain is a very common issue in a range of MDs and variations on pain prevalence across diagnostic groups appears to be minor. Generally, these people (especially those with FSHD and DM) experience moderate pain on average, which most frequently occurs to body areas such as lumbar spine, shoulders and legs. Our review also highlights the fact that chronic pain negatively impacts these individuals' daily life activities and may be also associated with the decreased quality of life among people with MDs.

The estimated prevalence of chronic pain appears to be similar across diagnostic groups: 68% (95% CI: 52%−82%) in FSHD, 65% (95% CI: 51%−77%) in DM, 62% (95% CI: 50%−73%) in BMD/DMD, and 60% (95% CI: 48%−73%) in LGMD, with no statically significant differences detected. However, it should be noted that heterogeneity was high in some diagnostic groups, reducing the certainty of these findings.

Importantly, the prevalence figures observed are considerably higher than the estimates of the prevalence of chronic pain in the general population,[17,24,50] and seem to be comparable to that of some other neurological conditions resulting in physical disabilities, such as cerebral palsy (with an overall mean pain prevalence of 70% (95% CI: 62%−78%) in adult patients)[52] and multiple sclerosis (an estimated pooled overall pain prevalence was 63% (95% CI: 55%−70%).[20]

While FSHD and DM are the most frequently investigated disorders, information of BMD/DMD and LGMD is relatively limited, with the remaining diagnostic groups including CMD, EDMD, OPMD, and Distal MDs barely explored.

In people with FSHD, a high number of studies (n = 8) allowed us to explore possible sources of heterogeneity in prevalence estimates. Of interest, the prevalence rate of chronic pain in FSHD appeared to decrease from a pooled estimate of 79% (95% CI: 74%−83%) in the 2000s to 57% (95% CI: 35%−78%) in the 2010s. Due to the lower number of studies published in recent years and the significant heterogeneity among these studies, we cannot be confident that a true reduction of chronic pain prevalence has occurred over time. However, over the past 10 years, the field of neuromuscular disorders has gained increased attention from health professionals and researchers, and evidence-based guidelines have been developed to advise standards of care and management of patients with FSHD.[2,46,47] Adoption of evidence-based guidelines and improvement in long-term care and management of the disorder, may have resulted in a reduced prevalence of chronic pain. Furthermore, the recognition of chronic pain in FSHD in the 2000s[25,31,32,48] may have led to better recognition of pain as an important problem in people with FSHD among health professionals, and, subsequently, improved pain management that may also have contributed to a decreasing prevalence in the following decade.

There is some evidence of geographical variation regarding chronic pain prevalence in FSHD, with the highest pooled prevalence estimate in America (82%, 95% CI: 76%−88%), followed by France (75%, 95% CI: 63%−85%) and the UK (56%, 95% CI: 51%−60%). If this geographical variation were true, it may be ascribed to genetic, cultural and/or lifestyle (eg, physical activity, obesity etc) differences across populations. However, it is not possible to draw conclusions at this stage, based on the limited data points and potential confounding factors including the publication date of studies, the methods of data retrieval and/or possible publication bias. Importantly, the studies from the UK reporting the lowest prevalence are coincidentally also the newer studies. As subgroup analysis by publication date suggested a decreased prevalence over time in FSHD, this is a confounding factor that could at least partly explain the geographic variations observed. Second, the two studies conducted in the UK were both retrospective analyses. One analyzed patient-reported data obtained from the UK FSHD Patient Registry,[39] while the other reviewed clinical records of children with FSHD in a single pediatric neuromuscular disorder centre.[45] As mentioned by both articles, information on pain was not always available for each participant. Thus, it is possible that the UK studies provide an underestimate of the true population prevalence. Finally, the possibility of publication bias was identified for prevalence estimates in FSHD, with the LFK index suggesting that studies with a higher prevalence of chronic pain were more likely to be published. Thus, it is possible that publication bias is at least partly responsible for the greater prevalence of chronic pain observed in people with FSHD in the USA and France.

As reported by Morís, Wood, FernáNdez−Torrón, González Coraspe, Turner, Hilton−Jones, Norwood, Willis, Parton, Rogers, Hammans, Roberts, Househam, Williams, Lochmüller, Evangelista, FernáNdez-Torrón, Hilton-Jones,[39] chronic pain tends to be more prevalent in women with FSHD, in line with the findings of other systematic reviews examining pain prevalence in the general population [26,35,37]. This is an interesting finding because gender influences the clinical expression of FSHD in an opposite way, with males normally presenting with an earlier onset of motor impairment and more severe disability.[43] The underlying mechanisms of gender variations in pain are still unclear, although some researchers have suggested that this phenomenon may be associated with socio-cognitive factors, the impact of sex hormones or endogenous opioid function.[5,6,18] Evidence of possible specific mechanism in FSHD contributing to gender difference in pain requires further exploration.

Chronic pain intensity appears to range from mild to severe across people with FSHD and DM. The average pain intensity was typically moderate. Our findings indicated that there was no significant difference on average pain intensity between DM and FSHD. However, as suggested by Jensen, Abresch, Carter, McDonald,[31] who attempted to evaluate chronic pain in a number of diagnostic groups of MDs, people with DM seem to more frequently experience severe pain than other MDs such as FSHD and LGMD. The underlying reasons for this remain unknown. The distinct nature of DM with predominant myotonia, myalgia, and multisystemic involvement[1] may contribute to the high frequency of severe chronic pain. Studies that assess pain intensity and severity in BMD/DMD and LGMD are very limited.

Lower back, shoulder and legs are the most frequent sites of chronic pain among people with FSHD, DM, BMD/DMD, and LGMD. In general, these localities reflect the body areas that are most commonly affected by specific MDs (see Appendix A).

No clear differences in pain quality across diagnostic group of MDs could be identified. "Tugging," "sharp," "dull," and "tiring" were the most common descriptors used. A subgroup of individuals described their pain as "radiating" (35%) and "burning" (35%), indicating the possibility of neuropathic pain in some forms of MDs.[40] However, to diagnose neuropathic pain and distinguish it from nociceptive pain, a more standardized diagnostic assessment is needed.[4] As the underlying pathophysiological mechanisms[54] and some first line treatments[23] are distinct, the differentiation between neuropathic pain and nociceptive pain should be further explored in future studies as it may be critical for planning effective pain mechanism-based treatment(s) in people with MDs.

Chronic pain has a negative impact on activities of daily living in people with MDs, and may also contribute to decreased quality of life. According to the findings from the BPI, occupational and domestic activities, recreational activities and mobility are daily life domains with the highest levels of pain interference in people with MDs, similar to the findings from the survey of the general population in Europe that chronic pain seriously affects the various domains of daily and working lives.[8] Further research is required examining the effects of chronic pain on sleep, mood, and QoL in MDs. Notably, features of chronic pain (eg, intensity, location, quality) with the greatest impact on these outcomes have barely been explored, and would be important to include in future studies.

There are several limitations that should be considered when interpreting our findings. First, the studies included in this review were most targeted at the adult population with FSHD and DM and were largely conducted in Europe and the USA. As such, information on the nature and scope of chronic pain among other diagnostic groups, in other regions of the world, and among children and teenagers is limited. Secondly, the tests for statistical heterogeneity among estimates of pain prevalence in FSHD and DM demonstrated high variability between studies. There were limited opportunities to explore sources of this variability because of the insufficient number of studies for comparison and a lack of variability in the characteristics of the included studies. Overall, the risk of bias of the included studies in this review was moderate to low. However, nearly half of the included studies had high risk of selection bias and nonresponse bias. Five of the 12 included studies recruited their participants from either an NMD clinic or training centre, and researchers were often deemed to use an inappropriate sampling method (a convenience sample) while selecting participants. Moreover, one study[51] only included FSHD participants with the ability to walk independently and defined levels of strength in specific muscle groups, which is not representative of the target population. Response rates of the survey studies (ranging from 45% to 78%) were generally low, with only one survey[32] matching the criteria of low-risk response rate (≥75%) based on the appraisal tool utilized to assess study quality.[29] Nevertheless, none of these studies performed an analysis to compare relevant demographic characteristics between responders and non-responders, and thus, may have been exposed to high risk of nonresponse bias. Furthermore, the possibility of publication bias was detected concerning estimates of pain prevalence in FSHD, with studies reporting high pain prevalence more likely to be published. This may explain part of the high heterogeneity presented across estimates of FSHD. Thirdly, due to limitations in the number of studies and outcome measures utilized, this review was not able to provide a clear appreciation on the most common causes of pain problems or whether different phenotypes of chronic pain (eg, neuropathic vs nociceptive pain) exist in people with MDs. Missing data may also be an important issue. Ten studies did not report independent data on individual diagnostic groups involved or did not fully report the outcomes of interest. We contacted the authors of these studies, but unfortunately, we were unable to obtain the data as requested owing to either no response or a response with data not available, so we had to exclude them. Finally, the analysis strategy was not preregistered on PROSPERO due to uncertainty regarding the availability of data for a specific synthesis method.

Conclusion

Chronic pain affects a high proportion of people across a range of MDs, with many describing their pain as frequent, widespread and severe. Our findings strengthen the evidence that routine pain assessment and management should be part of the standards of care in people with MDs. Future epidemiological studies that focus on chronic pain in lesser explored MDs, regions outside the USA and Europe and younger age groups are needed.

The raw data and all analysis functions of this study can be accessed through Mendeley Data http://dx.doi.org/10.17632/cfbskh2xn8.2.

Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.jpain.2021.04.001.

Appendix A Prevalence and clinical characteristics of MDs by diagnoses

Diagnoses	Prevalence	Age of onset	Gender	Muscle affected	Other systems affected	Cognitive deficits	Life expectancy
DMD	1/3,500	Aged three to five years	Males	Proximal muscles of the arms and legs; calf hypertrophy, muscle fibrosis, contractures	Respiratory or cardiac failure in late stage	Mild to severe	Reduced; late teens to early twenties
BMD	1/30,000	Around 12 years	Males	Proximal muscles of the arms and legs; calf hypertrophy, calf pain, cramps	Dilated cardiomyopathy	Mild	Reduced; the fourth or fifth decade
DM	1/8,000	DM1:Variable from birth through to adulthood DM2: Adulthood	Males/ females	DM1: Cranial, trunk, and distal limb muscles; myotonia DM2: Proximal limb muscles; myalgia	Cardiac conduction defects, tachyarrhythmias, gastrointestinal symptoms, cataracts, hyperglycemia, brain abnormalities	Mild	DM1: reduced DM2: almost normal
FSHD	1/20,000	late childhood or adolescence	Males/ females	Facial, shoulder, leg, thigh, and pelvic girdle muscles; scapular winging	Hearing loss and retinal telangiectasias	Not common	Most normal
LGMD	1:15,000 (autosomal recessive types)	Childhood or adolescence or adulthood.	Males/ females	Pelvic or shoulder girdle muscles	Respiratory or cardiac involvement depends on forms	Not common	Most normal
CMD	Unknown in general	In the uterus or in the first year of life	Males/ females	Generalized hypotonia and muscle weakness	Brain malformation and visual loss depend on forms	Severe in FCMD	Reduced in some forms
EDMD	1/100,000	Late childhood or adolescence	Males/ females	Proximal in the upper limbs and distal in the lower limbs; contractures of the elbows, Achilles tendons and cervical extensors	Cardiomyopathy	Not common	Reduced; middle age
OPMD	1/100,000	The fourth to sixth decade	Males/ females	Extraocular muscle paralysis and mild proximal muscles weakness; dysphagia and dysphonia	Superior visual field defect	Not common	Normal
Distal MD	Unknown in general	Early adulthood to middle age	Males/ females	Predominant weakness in the feet and/or hands	Not common	Not common	Normal

DMD, Duchenne muscular dystrophy; BMD, Becker muscular dystrophy; DM, Myotonic muscular dystrophy; FSHD, Facioscapulohumeral muscular dystrophy; LGMD, Limb-girdle muscular dystrophy; CMD, Congenital muscular dystrophy; FCMD, Fukuyama- Congenital muscular dystrophy; EDMD, Emery-Dreifuss Muscular Dystrophy, OPMD, Occulopharyngeal Muscular Dystrophy; Distal MD, Distal Muscular Dystrophies

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